**Harsh Kumar** | **200020056**
**Chemical Engineering** | **B.Tech.**
**Indian Institute of Technology Bombay** | **Gender: Male**
**Contact: harshkrjha9949@gmail.com | 6201001984** | **DOB: 06/03/2002**

| Examination | University | Institute | Year | CPI / % |
|---|---|---|---|---|
| Graduation | IIT Bombay | IIT Bombay | 2024 | 7.53 |
| Intermediate | CBSE | Bijendra Public School | 2019 | 86.00% |
| Matriculation | CBSE | Bijendra Public School | 2017 | 10 |

Pursuing a Minor Degree in the Department of **Computer Science and Engineering** at IIT Bombay

## Professional Experience

**Software Development Intern** | Hilti Technology Solutions India *(May '23 - Jul '23)*
*Received a **Pre-Placement Offer** based on exceptional performance during the internship*
- Integrated image detection in the app to automate product selection, reducing **10+** tedious user interventions
- Created new firestop selection feature leveraging **Xamarin** and C# in **.NET** to connect AI logic with Interface
- Enhanced system stability by migrating **13+** APIs from **Node.js** to **ASP .Net** collaborating with global team

**AR-VR Game Development Intern** | Mirana Innovations *(May '22 - Jul '22)*
*Startup with a goal to make good design, aesthetics, and ergonomics with reliable technology in the toy industry*
- Researched through AR libraries including **Wikitude**, **Vuforia** and **AR Foundation** to find optimal solution
- Engineered **MVVM** styled **C# SDK** for tranferring data to host application and handling the **vehicle state**
- Developed an advanced **AR-enabled** remote console for driving mini toy cars within a virtual space overlay

**Software Development Intern** | Gupshup *(Apr '21 - Jul '21)*
*A **Unicorn startup** working towards better customer engagement through **Conversational AI** messaging*
- Developed **3+** chatbot models using **Natural Language Processing** (NLP) and **Node.js** on Gupshup's IDE
- Designed and developed smart contextual AI assistants and chatbots using **RASA Development Framework**
- Deployed CI/CD pipelines to automate ChatBot service using REST APIs; trained it to increase performance

## Key Projects

**InstiApp** (**11,000+** Users & **2,500+** MAU) | Developers' Community, IIT Bombay *(May '21 - Apr '23)*
- Deployed Community building feature on app providing Institute Bodies access to create and manage forums
- Ideated the creation and launch of flutter based app, onboarding **500+ iOS users** for **1st** time in app's history
- Upgraded the app with Live Posts using Angular-Django backend, collaborating with the Institute Media Body

**New Application Service Centre** (**15,000+** Stud. & **3,000+** Employees) | DevCom IITB *(Jan '23 - Apr '23)*
- Implemented and showcased a unified base frontend template for all IITB web applications, including **ERP**
- Used **Vue** and moved to **Single - Spa** creating user-friendly portal, connecting to JAVA based static backend
- Enhanced mobile website usability through research-driven design teaming with the **Institute's ASC Head**

**DC Print** | Developers' Community, IIT Bombay *(Jan '22 - Apr '22)*
- Ideated and developed a portal to simplify remote printing for more than **1K** students using network printing
- Used **C's CUPS** library and **sub-processes in Python** to develop portal to send requests to universal printer
- Worked in a team of **2** to manage queue of multiple requests in DC-Print and print with options at frontend

**Resobin** | Developers' Community, IIT Bombay *(Oct '21 - Dec '21)*
- Contributed to development of "**ResoBin**" (Course resources portal of IITB) used by **7K+** enrolled students
- Developed the most used feature 'Minors' info page utilising **Django Rest** Framework for backend integration
- Organised and helped team in feature development required and increased user engagement by improving **UX**

**Digital Notice Board** | Developers' Community, IIT Bombay *(May '21- Sep '21)*
- Implemented a sustainable approach towards a paper-free campus by eliminating physical event announcements
- Utilised **Raspberry Pi** as a hardware platform to develop and deploy **Python** scripts using relevant libraries
- Implemented logic to connect **realtime** content to InstiApp updating the live display feed in regular interval

**Operating System** | Course Projects | Prof. Mythili Vutukuru *(Jan '22 - Apr '22)*
- Programmed **C++ algorithms** for process classification, memory management, **pthreads synchronization**
- Implemented file-system management on **MIT** developed open-source **XV6** operating system adding modules

## Other Projects

**Supervised Industrial Development Program** | Aarti Industries | Course Project *(Jul '23 - Present)*

- Partnering with industry experts to tackle a crucial challenge: eliminating cresol from nitrated toluene streams
- Employing simulations(**DWSIM**) & experimenting, improving process resulting in daily savings of **10 lakh**

**Optimisation of the Geometry of an Orifice Meter** | Course Project | Prof. Khakhar *(Mar '22 - Apr '22)*

- Collaborated with 2 peers to optimise the geometry of a standard orifice meter to reduce the energy consumption
- Generated 2-D finite element mesh on **Gmsh**, used computational strategy to simulate flow using **OpenFOAM**

**Optimising Water Storage Tanks** | Course Project | Prof. P. Sunthar *(Jan '22 - Apr '22)*

- Optimised water tank to keep temperature optimal in high solar radiation and sub-zero temperature areas
- Contributed with 1 peer in a team of **8** to solve the problem by computational analysis using **OpenFoam**
- Run simulation of **10 hours** and observing the temperature change and optimising parameters to get results

**White December** | **Podcast on SPOTIFY** | Self Project *(Oct '20 - Apr '21)*

- Managed a three-member team to direct and produce a four-episode Podcast, telling emotional fictional story
- Provided Voice-Over and wrote the story for the podcast for around **1K average listens** for 7 episodes written

## Technical Skills

| | |
|---|---|
| **Languages** | C, C++, C#, Python, JavaScript, HTML, CSS, LaTeX, Dart, SQL, Java |
| **Libraries & Frameworks** | Django, PyTorch, NumPy, Pandas, Scikit-learn, Flutter, Angular, Xamarin, .NET, Wikitude, Vuforia, Vue.JS |
| **Softwares & Others** | MySQL, MongoDB, Ubuntu, Linux, NodeJS, Android Studios, Git, Firebase, MATLAB, Unity, XCode |

## Positions of Responsibility

**Project Lead** | DevCom, IIT Bombay *(May '22 - Apr '23)*

*One of the biggest Developers' Communities in India for design, development & maintenance of technical projects*

- Led a team of **21** student developers tasked with maintaining **critical digital infrastructure** for the institute
- Responsible for enhancing, managing, and expanding the user base of DevCom projects with robust support
- Increased the user base of InstiApp by **10%** by forming a **4**-member design team to improve design & **UI/UX**

**Institute Student Mentor** | Student Mentorship Program *(May '22 - Present)*

*Platform for enabling constructive and positive interaction among junior and senior students*

- Leading **11** mentors, improving support for groups of **12** students each, fostering resilience and effective coping
- Part of a **143**-membered team aiming to play a facilitative, supportive and developmental role for new entrants
- Ensuring that the mentees experience a smooth transition into the new academic and social life at IIT Bombay

**Academic Mentor** | Department of Chemical Engineering *(May '22 - Apr '23)*

*A team of **40** members for student mentorship program to build a support system for **160+** students*

- Selected into a team of **40** members out of **98** applicants, based on SOP, ethics and extensive peer reviews
- Mentored **6** sophomores, fostering support to manoeuvre academic load and actively address their social needs
- Involved in collecting and co-authoring extensive course reviews for over **60** courses for the D-AMP blog

**Teaching Assistant** | Computer Systems BootCamp *(May '22 - Jun '22)*

*An initiative of Computing Systems Education and Research Lab, IIT Bombay*

- Worked in a team of **17** TAs under **3** instructors to ideate and execute bootcamp for students all over india
- Helped with creation of content (Assignment code) to equip **50+** participants with basic coding knowledge
- Conducted tutorial sessions building an HTTP **application server** capable of executing custom user functions

## Extracurricular Activities

- Wrote and directed '**Chor**', a short-film which won the **1st** position in Inter-IIT Culturals Meet, Madras *(2023)*
- Created an advertisement for **Xiaomi** Ind's ad-making competition which won the **first** prize nationwide *(2023)*
- Scripted a street play, based on youth's political ignorance, which won **1st** position in Street Play GC *(2023)*
- Stood **second** among **100+** participants in Inter-Hostel Kavya Sandhya Poetic General Championship *(2022)*
- Played **lead** in the radio play **Mahdood** which stood **3rd** in **Sarang-IITM**, promoting inclusivity *(2020)*
- Wrote **a satirical play** performed in Annprod, the annual theatre showcase before **500+** audience *(2022)*
- Secured **3rd** position in **Aeronautical GC** for programming PlutoX drone to avoid obstacles *(2021)*
- Played the character of a police man in **Institute General Championship** radio play for Hostel 15 *(2020)*